



A prospective study of the prostate health index density and multiparametric magnetic resonance imaging in diagnosing clinically significant prostate cancer

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Purpose: To evaluate the predictive performance of the prostate health index (PHI) and PHI density (PHID), for clinically significant prostate cancer (csPCa) in patients with a PI-RADS score ≤ 3 .

Materials and Methods: Patients tested for total prostate-specific antigen (tPSA, ≤ 100 ng/mL), free PSA (fPSA), and p2PSA at Peking University First Hospital were prospectively enrolled. Possible predictive factors of csPCa were analyzed using the receiver operating characteristic (ROC) curve. Results were expressed as area under the curve (AUC) with 95% confidence intervals (CI). The cutoff values of PHI and PHID were determined.

Results: We enrolled 222 patients in this study. The prevalence of csPCa in the PI-RADS ≤ 3 subgroup ($n=89$) was 22.47% (20/89). Age, tPSA, F/T, prostate volume, PSA density, PHI, PHID, and PI-RADS score were significantly associated with csPCa. PHID (AUC: 0.829 [95% CI: 0.717–0.941]) was the best predictor of csPCa. PHID >0.956 was set as the threshold of suspicious csPCa with a sensitivity of 85.00% and a specificity of 73.91%, avoiding 94.44% of unnecessary biopsies but missing 15.00% csPCa. A threshold of PHI ≥ 52.83 showed the same sensitivity but a rather lower specificity of 65.22% that avoided 93.75% of unnecessary biopsies.

Conclusions: PHI and PHID have the best predictive performance of csPCa in patients with PI-RADS score ≤ 3 . A threshold value of PHID ≥ 0.956 may be used as the criterion for biopsy in these patients.

Keywords: Biomarker; Clinical laboratory techniques; Diagnostic imaging; Multiparametric magnetic resonance imaging; Prostate cancer

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INTRODUCTION

Prostate cancer (PCa) is the most common cancer among

men [1]. Prostate-specific antigen (PSA) has been widely used for PCa screening. However, given the rather low specificity of PSA, overdiagnosis and overtreatment have become

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issues of concern. Furthermore, other conditions such as benign prostate hyperplasia (BPH) or urological procedures may also lead to a rise in the PSA level [2]. Other blood or urine based biomarkers, such as prostate-specific membrane antigen and prostate cancer antigen 3 (PCA3), as well as multivariate models have been discovered and developed [3,4]. Discovered in 2000, a precursor form of PSA called [-2] proPSA (also known as p2PSA), is known to be associated with PCa and can help distinguish PCa from BPH because of its high specificity for PCa [5,6]. Prostate health index (PHI), a mathematical combination of serum concentrations of p2PSA, free PSA (fPSA), and total PSA (tPSA) was developed by Beckman Coulter. It has a better PCa prediction performance, especially for patients with PSA level 2–10 ng/mL.

Multiparametric magnetic resonance imaging (mpMRI) is a useful and precise method to evaluate probable prostatic lesions, and it provides orientations for further workups such as prostate biopsy [7,8]. Lesions discovered on mpMRI have been assessed using the Prostate Imaging Reporting and Data System (PI-RADS) version 2.1 [9] and categorized based on their likelihood for PCa. Per clinical experience, a few borderline lesions are often scored 3 (named “equivocal lesions”), indicating an undetermined probability of PCa. Nevertheless, some non-suspicious lesions (i.e., PI-RADS score 1 or 2) are finally diagnosed as cancers with time. Therefore, a deeper classification of non-suspicious or equivocal lesions may further lower the proportion of false negative PCa. The present study aimed to evaluate the predictive performance of PHI and PHI density (PHID) for PCa in patients with PI-RADS scores ≤ 3 and to determine whether PHI and PHID can help guide the decision to perform a biopsy for these patients.

MATERIALS AND METHODS

1. Study population

Patients who were tested for tPSA, fPSA, and p2PSA at Peking University First Hospital between July 2017 and October 2021 were prospectively enrolled with their information recorded. Overall, 399 consecutive patients were enrolled. All the patients were Asian men. The exclusion criteria were tPSA >100 ng/mL, absence of biopsy, tumor other than prostate adenocarcinoma on biopsy, values exceeding the maximum limit of detection, mpMRI with b-value $<1,400$ s/mm² that cannot be used for PI-RADS v2.1 scoring, and insufficient clinical data. Ultimately, 222 patients were included (Fig. 1). This study protocol was reviewed and approved by the Ethics Committee of Peking University First Hospi-

tal (protocol code 2016-1275). The need for informed consent was waived by the Ethics Committee of Peking University First Hospital. All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

2. Clinical outcome

All patient data were collected (clinical and pathological characteristics). Prior to biopsy, age, and body mass index information was measured and/or recorded for each patient. Serum tPSA, fPSA, and p2PSA levels were also tested. The free-to-total ratio of PSA (F/T) was calculated by dividing tPSA by fPSA. The prostate volume (PV) was calculated using the formula width $\times$ length $\times$ height $\times 0.52$ with the parameters measured on mpMRI. PSAD was calculated by dividing tPSA by PV. PHI was calculated using the following formula:

$$PHI = \frac{p2PSA (pg/mL)}{fPSA (ng/mL)} \times \sqrt{tPSA (ng/mL)}$$

PHID was calculated by dividing PHI by PV.

3. The mpMRI technique

Baseline prostate mpMRI was performed on a 3-T scanner (Discovery[®] HD 750; GE Healthcare) using a phased-array coil. The protocol used in this study included b-values of 0, 800, and 1,400 s/mm² that met the recommended minimal requirements for PI-RADS v2.1. Details of the mpMRI parameters are presented in Supplementary Table 1.

4. Image interpretation

The images were prospectively interpreted by a collaboration group of experienced radiologists and urologists. PI-RADS v2.1 structured criteria developed by the European Society of Urogenital Radiology [9] was applied and scores were given to each patients.

5. Biopsy procedure

All included patients underwent prostate biopsy. A total of 218 patients received transrectal ultrasound-guided systematic biopsy and 149 of them also underwent a concomitant cognitive mpMRI-targeted biopsy, i.e., extra cores were obtained from suspicious regions found on mpMRI, and these cores were specifically labeled. Four patients received only targeted biopsy. A group of experienced genitourinary pathologists evaluated all the specimens to determine the diagnosis and Gleason score in positive cases. Clinically sig-

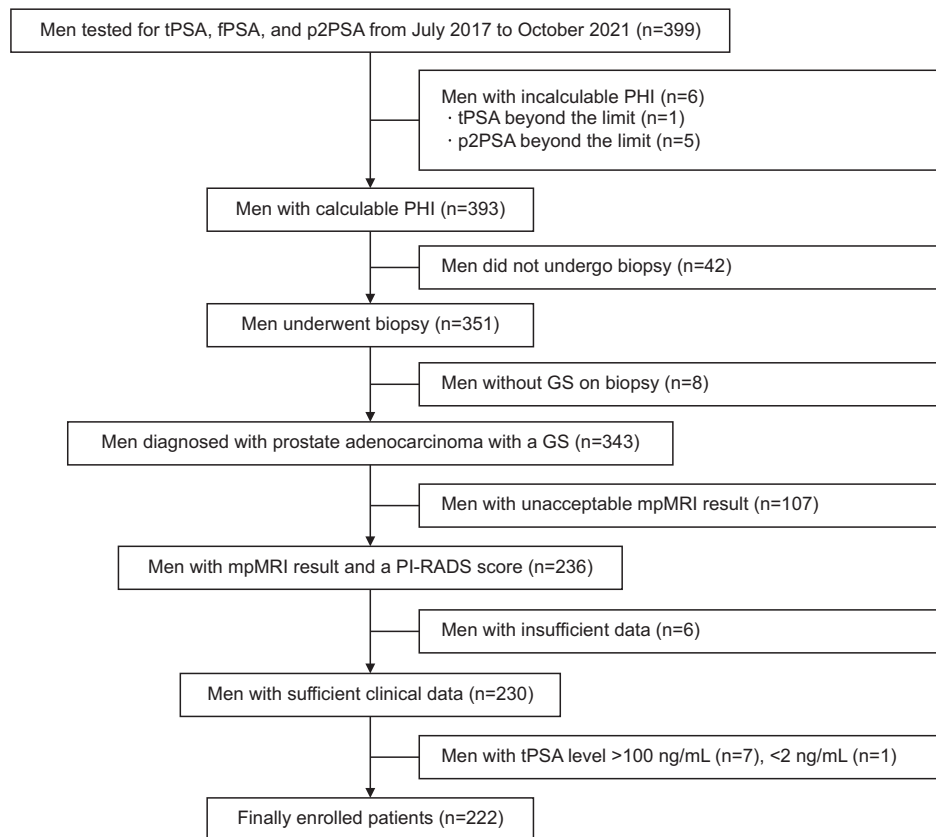


Fig. 1. The inclusion and exclusion flow chart of the study cohort. tPSA, total prostate-specific antigen; fPSA, free PSA; PHI, prostate health index; GS, Gleason score; mpMRI, multiparametric magnetic resonance imaging; PI-RADS, Prostate Imaging Reporting and Data System.

nificant prostate cancer (csPCa) was defined as having a Gleason score 3+4, 4+3, 8, 9, or 10 (the Epstein criteria [10]). PCa, including csPCa and insignificant PCa, was diagnosed if malignancy was detected in any biopsy core.

6. Statistical analysis

Statistical analysis was performed using Python 3.7.6 with statsmodels 0.11.0, scipy 1.4.1, and sklearn 0.22.1 unless specified otherwise. Statistical significance was set at $p < 0.05$, and all analyses were two-sided.

The primary endpoint of the study was the identification of csPCa on biopsy. The association between the possible risk factors and the presence of csPCa or PCa was investigated using univariate analyses. Continuous variables were compared using Student's *t*-test and the Mann–Whitney *U* test, and categorical variables were compared using the Fisher's exact test or chi-square test, as deemed appropriate. Multivariate logistic regression analysis was conducted to identify independent predictors of csPCa, as well as build combined models of carefully selected individual factors. Independent predictors and combined models were analyzed using the receiver operating characteristics (ROC) (through the estimation of the area under the curve, AUC) to evaluate their diagnostic values for csPCa. The DeLong test was used to compare AUCs. The one with largest Youden index

was selected as the most appropriate cutoff value. The decision curves were plotted using R 4.2.3 after the output of predictive probabilities of factors/models calculated using Python 3.7.6 with statsmodels 0.11.0.

RESULTS

In all 222 patients, 100 (45.05%) had gray-zone PSA level (2–10 ng/mL). Among them, 44 (44.00%) had csPCa and 56 (56.00%) had PCa. The mean age ($p = 0.001$), PSAD ($p < 0.001$), PHI ($p < 0.001$), PHID ($p < 0.001$), and PI-RADS v2.1 score ($p < 0.001$) of patients with csPCa were significantly higher than those of patients without csPCa (Table 1). A significantly lower PV ($p < 0.001$) and F/T ($p = 0.006$) were associated with csPCa (Table 1). In patients with PSA level 10–20 ng/mL ($n = 72$, Supplementary Table 2), above 20 ng/mL ($n = 50$, Supplementary Table 3) or the whole cohort (Supplementary Table 4), higher PSAD, PHI and PHID as well as lower F/T and PV were associated with csPCa, showed a similar pattern of those with gray-zone PSA level (the currently recommended PSA level for PHI testing).

Notably, in patients whose lesions were equivocal or negative on mpMRI (i.e. PI-RADS score ≤ 3 , $n = 89$), 20 patients (22.47%) had csPCa with a significantly higher age ($p = 0.005$), PSAD ($p < 0.001$), PHI ($p = 0.001$) and PHID ($p < 0.001$), mean-

Table 1. Patient characteristics of subgroups

Variable	PSA 2–10 ng/mL					PI-RADS score 1–3				
	csPCa		PCa		Total	csPCa		PCa		p-value
	YES	NO ^a	YES	NO		YES	NO ^a	YES	NO	
Patients	100 (45.05)	44 (44.00)	57 (57.00)	43 (43.00)	89 (40.09)	20 (22.47)	69 (77.53)	31 (34.83)	58 (65.17)	-
Age (y)	65.09±7.35	67.73±6.93	67.18±6.48	62.33±7.52	64.63±6.97	68.45±6.49	63.52±6.71	66.71±6.90	63.52±6.76	0.040
BMI (kg/m ²)	24.46±3.10	24.95±3.23	24.61±3.13	24.27±3.04	24.60±2.87	25.39±2.93	24.37±2.81	24.92±2.72	24.43±2.94	0.452
tPSA (ng/mL)	6.71	6.75	6.67	6.81	9.77	10.13	9.75	10.10	9.76	0.360
	(5.17–8.29)	(5.14–8.56)	(5.18–8.43)	(5.05–7.95)	(6.62–15.28)	(6.66–15.35)	(6.62–15.16)	(6.19–15.09)	(7.12–15.33)	
F/T	0.14	0.12	0.13	0.15	0.14	0.09	0.15	0.10	0.15	0.027
	(0.09–0.20)	(0.08–0.17)	(0.09–0.18)	(0.11–0.20)	(0.09–0.21)	(0.08–0.15)	(0.10–0.21)	(0.08–0.19)	(0.11–0.20)	
PV (mL)	44.27	36.80	38.00	48.93	58.16	38.28	65.50	46.22	66.88	<0.001
	(29.41–63.27)	(26.90–47.81)	(28.00–53.00)	(38.75–69.78)	(41.50–80.36)	(24.63–50.55)	(48.78–87.05)	(27.13–58.72)	(49.47–94.31)	
PSAD (ng/mL ²)	0.14	0.19	0.17	0.11	0.17	0.42	0.15	0.31	0.15	<0.001
	(0.09–0.23)	(0.11–0.29)	(0.11–0.26)	(0.09–0.15)	(0.11–0.29)	(0.18–1.10)	(0.10–0.24)	(0.16–1.10)	(0.10–0.20)	
PHI	53.04	67.75	65.53	34.83	47.47	75.33	43.02	65.17	43.57	0.010
	(33.36–72.68)	(57.81–89.23)	(50.94–86.93)	(29.61–46.18)	(30.59–72.36)	(54.04–92.70)	(29.79–61.57)	(36.73–84.71)	(30.08–62.24)	
PHID	1.12	1.94	1.78	0.75	0.74	1.88	0.65	1.24	0.66	<0.001
	(0.68–2.06)	(1.15–3.23)	(0.96–2.95)	(0.44–1.15)	(0.44–1.26)	(1.11–3.73)	(0.42–1.01)	(0.67–3.01)	(0.43–0.92)	
PI-RADS v2 score										
1	2 (2.00)	0 (0.00)	0 (0.00)	2 (4.65)	7 (7.87)	0 (0.00)	7 (10.14)	1 (3.23)	6 (10.34)	0.006
2	17 (17.00)	2 (4.55)	3 (5.26)	14 (32.56)	29 (32.58)	2 (10.00)	27 (39.13)	4 (12.90)	25 (43.10)	
3	26 (26.00)	7 (15.91)	12 (21.05)	14 (32.56)	53 (59.55)	18 (90.00)	35 (50.72)	26 (83.87)	27 (46.55)	
4	40 (40.00)	24 (54.55)	29 (50.88)	11 (25.58)	-	-	-	-	-	
5	15 (15.00)	11 (25.00)	13 (22.81)	2 (4.65)	-	-	-	-	-	
Gleason Grade Group										
Not cancer	43 (43.00)	-	-	43 (100.00)	58 (65.17)	-	58 (84.06)	-	58 (100.00)	-
1	13 (13.00)	-	13 (22.81)	-	11 (12.36)	-	11 (15.94)	11 (35.48)	-	
2	21 (21.00)	21 (47.73)	21 (36.84)	-	8 (8.99)	8 (40.00)	-	8 (25.81)	-	
3	11 (11.00)	11 (25.00)	11 (19.30)	-	5 (5.62)	5 (25.00)	-	5 (16.13)	-	
4	9 (9.00)	9 (20.45)	9 (15.79)	-	4 (4.49)	4 (20.00)	-	4 (12.90)	-	
5	3 (3.00)	3 (6.82)	3 (5.26)	-	3 (3.37)	3 (15.00)	-	3 (9.68)	-	

Values are presented as number (%), mean±standard deviation, or median (interquartile range).

csPCa, clinically significant prostate cancer; PCa, prostate cancer; BMI, body mass index; tPSA, total prostate-specific antigen (PSA); F/T, free-to-total PSA ratio; PV, prostate volume; PSAD, PSA density; PHI, prostate health index; PHID, PHI density; PI-RADS, Prostate Imaging Reporting and Data System.

^aPatients who were cancer-free or having insignificant cancer (i.e., positive cores ≥ 1 and Gleason score ≤ 6) were included in csPCa-negative subgroup.

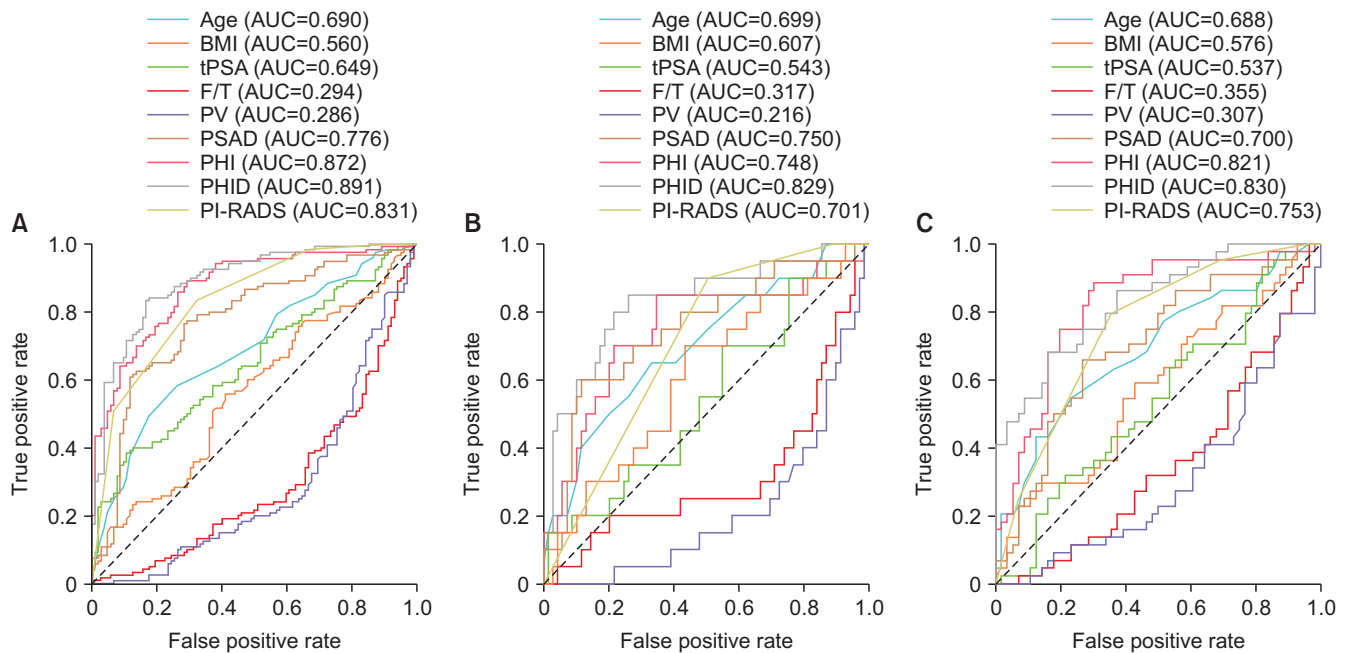


Fig. 2. Univariate ROC curve for csPCa prediction. (A) The whole cohort (n=222). (B) PI-RADS ≤ 3 subgroup (n=89). (C) Gray-zone PSA subgroup (n=100). ROC, receiver operating characteristic; csPCa, clinically significant prostate cancer; AUC, area under the curve; BMI, body mass index; tPSA, total prostate-specific antigen; F/T, free-to-total PSA ratio; PV, prostate volume; PSAD, PSA density; PHI, prostate health index; PHID, PHI density; PI-RADS, Prostate Imaging Reporting and Data System.

while their F/T ($p=0.007$) and PV ($p<0.001$) were significantly lower (compared to those who did not have csPCa) (Table 1).

The multivariate logistic regression model for csPCa was established as a function of age, tPSA, F/T, PV, PHI, and PI-RADS score (Supplementary Table 5). All variables but tPSA and F/T remained significantly associated, suggesting that these variables could be independent predictors for the diagnosis of csPCa on biopsy in all patients. For the subgroups, in the multivariate logistic regression model analysis for csPCa in association with age, F/T, PV, PHI, and the PI-RADS score, only F/T was not significantly associated. PHI and PHID significantly increased the predictive performance of multivariate combined model (age+tPSA+ F/T+PV+PHI+PI-RADS score) in all patients (PHI, $p=0.025$; PHID, $p=0.045$) but it showed no significant difference in the PI-RADS ≤ 3 subgroup (PHI, $p=0.165$; PHID, $p=0.493$) and the gray-zone PSA subgroup (PHI, $p=0.236$; PHID, $p=0.144$) (combined model: age+F/T+PV+PHI+PI-RADS score), as shown in Supplementary Table 6. PHID showed the best diagnostic accuracy (Fig. 2, Table 2) in the whole cohort (AUC=0.891, 95% confidence interval [CI]: 0.849–0.933, $p<0.05$). This was also the case in the PI-RADS ≤ 3 subgroup (AUC=0.829, 95% CI: 0.717–0.941, $p<0.05$) and the gray-zone PSA subgroup (AUC=0.830, 95% CI: 0.751–0.908, $p<0.05$). While adding PHI or PHID to each individual factors (Table 2), it significantly increased the AUC of ROC curve of each factors in all patients, the gray-

zone PSA subgroup, and the PI-RADS ≤ 3 subgroup (except age+PHI, $p=0.102$; BMI+PHI, $p=0.054$; PSAD+PHI, $p=0.662$; PSAD+PHID, $p=0.076$ in the PI-RADS ≤ 3 subgroup).

The detailed diagnostic performance of PHI and PHID was analyzed, as shown in Table 3. Cutoff values of PHI and PHID for csPCa were determined using the Youden index. For patients with a non-suspicious or equivocal mpMRI result, PHI ≥ 51.84 or PHID ≥ 0.956 (sensitivity=85.00%) was the best criterion for csPCa prediction. The missing rate using this criterion was 15.00%. However, PHID presented with a higher specificity (73.91%) and a higher negative predictive value (NPV) of 94.44% than PHI (specificity=65.22% and NPV=93.75%), and this could reduce the rate of unnecessary biopsies. For the gray-zone PSA subgroup, PHI ≥ 48.68 had a higher sensitivity of 88.64% and a higher NPV of 88.64% than the threshold PHID ≥ 1.42 (sensitivity=68.18% and NPV=77.05%). Notably, the criteria for csPCa prediction in all patients and patients with non-positive mpMRI results were associated with a significantly lower percentage of tumorous tissue in biopsy cores (Table 3). The decision curve analysis (DCA) showed PHI (except the PI-RADS ≤ 3 subgroup) and PHID had more benefit than other factors (Supplementary Fig. 1).

Table 2. AUC of ROC curve for csPCa prediction

Variable		All (n=222)	PI-RADS ≤3 (n=89)	Gray-zone PSA (n=100)
Age	Standalone	0.690 (0.621–0.759)	0.699 (0.563–0.835)	0.688 (0.582–0.795)
	Age+PHI	0.894 (0.852–0.936)	0.809 (0.697–0.920)	0.833 (0.753–0.914)
	p-value	<0.001	0.102	0.006
	Age+PHID	0.911 (0.873–0.949)	0.822 (0.707–0.938)	0.877 (0.809–0.944)
	p-value	<0.001	0.007	<0.001
BMI	Standalone	0.560 (0.484–0.635)	0.607 (0.465–0.749)	0.576 (0.462–0.690)
	BMI+PHI	0.873 (0.828–0.918)	0.754 (0.626–0.883)	0.832 (0.750–0.914)
	p-value	<0.001	0.054	<0.001
	BMI+PHID	0.898 (0.857–0.939)	0.830 (0.721–0.940)	0.846 (0.770–0.922)
	p-value	<0.001	0.001	<0.001
tPSA	Standalone	0.649 (0.577–0.720)	0.543 (0.395–0.690)	0.537 (0.422–0.652)
	tPSA+PHI	0.875 (0.830–0.921)	0.758 (0.625–0.891)	0.829 (0.746–0.912)
	p-value	<0.001	0.010	<0.001
	tPSA+PHID	0.892 (0.849–0.935)	0.825 (0.710–0.940)	0.834 (0.755–0.913)
	p-value	<0.001	0.001	<0.001
F/T	Standalone	0.294 (0.225–0.363)	0.317 (0.169–0.466)	0.355 (0.247–0.464)
	F/T+PHI	0.878 (0.832–0.923)	0.759 (0.612–0.907)	0.838 (0.756–0.921)
	p-value	<0.001	0.002	<0.001
	F/T+PHID	0.888 (0.844–0.932)	0.812 (0.685–0.940)	0.830 (0.752–0.909)
	p-value	<0.001	<0.001	<0.001
PV	Standalone	0.286 (0.217–0.356)	0.216 (0.105–0.326)	0.307 (0.203–0.412)
	PV+PHI	0.903 (0.863–0.942)	0.856 (0.758–0.953)	0.841 (0.765–0.917)
	p-value	<0.001	<0.001	<0.001
	PV+PHID	0.891 (0.848–0.933)	0.845 (0.743–0.946)	0.834 (0.755–0.913)
	p-value	<0.001	<0.001	<0.001
PSAD	Standalone	0.776 (0.712–0.839)	0.750 (0.619–0.881)	0.700 (0.595–0.805)
	PSAD+PHI	0.876 (0.831–0.921)	0.778 (0.647–0.908)	0.827 (0.743–0.911)
	p-value	0.001	0.662	0.034
	PSAD+PHID	0.890 (0.848–0.933)	0.840 (0.744–0.936)	0.828 (0.748–0.907)
	p-value	<0.001	0.076	0.002
PI-RADS	Standalone	0.831 (0.781–0.882)	0.701 (0.616–0.787)	0.753 (0.663–0.842)
	PI-RADS+PHI	0.903 (0.864–0.942)	0.827 (0.719–0.934)	0.853 (0.775–0.932)
	p-value	<0.001	0.002	0.006
	PI-RADS+PHID	0.908 (0.870–0.945)	0.864 (0.772–0.955)	0.861 (0.791–0.930)
	p-value	<0.001	<0.001	0.001
PHI		0.872 (0.826–0.918)	0.748 (0.610–0.886)	0.821 (0.736–0.907)
PHID		0.891 (0.849–0.933)	0.829 (0.717–0.941)	0.830 (0.751–0.908)

Values are presented as AUC (95% confidence interval).

AUC, area under the curve; ROC, receiver operating characteristic; csPCa, clinically significant prostate cancer; BMI, body mass index; tPSA, total prostate-specific antigen (PSA); F/T, free-to-total PSA ratio; PV, prostate volume; PSAD, PSA density; PHI, prostate health index; PHID, PHI density; PI-RADS, Prostate Imaging Reporting and Data System.

DISCUSSION

Prostate biopsy is the only confirmative procedure (to diagnose PCa) recommended for patients who have elevated PSA level and/or suspicious digital rectal examination (DRE) and/or abnormal imaging. Given the aggressiveness and

potential risks of biopsy (e.g., hemorrhage and infection), several triage strategies to avoid unnecessary biopsies have been studied. Attempts have been made using PSA related-indexes (F/T and PSAD), PI-RADS scores, and other biomarkers such as p2PSA and PCA3 [11]. To date, no consensus has been reached.

Table 3. Cutoff values of PHI and PHID for predicting csPCa

Group	Prevalence of csPCa (%)	Variate	Cutoff value ($\geq$)	Sensitivity (%)	Specificity (%)	Youden index	PPV (%)	NPV (%)	False negative rate (missed cases, %)	False positive rate (misdiagnosed cases, %)	Max% in core, median (IQR)		p-value
											Above cutoff	Below cutoff	
All (n=222)	54.05	PHI	60.37	89.17	70.59	0.5975	78.10	84.71	10.83	29.41	67.00 (66.50–90.00)	32.50 (10.00–80.00)	<0.001
		PHID	1.45	83.33	83.33	0.6667	85.47	80.95	16.67	16.67	67.00 (67.00–90.00)	50.00 (27.50–80.00)	<0.001
PI-RADS ≤ 3 (n=89)	22.47	PHI	51.84	85.00	65.22	0.5022	41.46	93.75	15.00	34.78	60.00 (34.00–80.00)	25.00 (7.50–32.50)	0.001
		PHID	0.956	85.00	73.91	0.5891	48.57	94.44	15.00	26.09	55.00 (29.50–80.00)	30.00 (7.50–32.50)	0.003
Gray-zone PSA (n=100)	44.00	PHI	48.68	88.64	69.64	0.5828	69.64	88.64	11.36	30.36	67.00 (50.00–90.00)	50.00 (20.00–75.00)	0.102
		PHID	1.42	68.18	83.93	0.5211	76.92	77.05	31.82	16.07	67.00 (60.00–85.00)	55.00 (31.00–80.00)	0.068

PHI, prostate health index; PHID, PHI density; csPCa, clinically significant prostate cancer; PPV, positive predictive value; NPV, negative predictive value; IQR, interquartile range; PI-RADS, Prostate Imaging Reporting and Data System; PSA, prostate-specific antigen.

Approved by the FDA in 2012 for PCa screening in men older than 50 years with normal DRE, and PSA levels between 4 and 10 ng/mL, PHI was shown to be the best predictor when compared with tPSA, PSAD, and PV [12]. A meta-analysis study done after showed that PHI had a higher diagnostic performance for PCa than %fPSA in patients with PSA levels between 2 and 10 ng/mL [13]. PHI also showed a better predictive performance of csPCa in patients with PSA level >10 ng/mL [6] and patients with prior negative biopsy results [14]. A PHI above the range 31–34 is considered to be associated with csPCa with a sensitivity of 80%–85% and a specificity of 38%–46% [15]. In a prospective Asian cohort study, a threshold PHI ≥ 36.96 with a sensitivity of 90% and a specificity of 521% had better performance than tPSA, to predict csPCa detection on biopsy. Combining PHI and PV, PHID was also reported to have a better performance [16,17]. A better predictive performance of PHI and PHID (than the other potential predictors for csPCa) was also shown in the present study.

Considering its higher diagnostic performance on PCa, when compared with tPSA, transrectal ultrasound and DRE, mpMRI has been used in the diagnostic workup of PCa and its role in pre-biopsy triage has been proven [18]. Studies have shown the synergistic effect of both mpMRI and PHI for csPCa detection [19]. PI-RADS, a classification system associating imaging results and likelihood of PCa, is scored from 1 to 5. Scores 1 and 2 are considered to be non-suspicious of cancer with a NPV of approximately 90%, but there is still a considerable chance of missing PCa cases. Scores 4 and 5 are considered to be suspicious for cancer (40% to 80% of patients with these scores are definitely diagnosed with csPCa) [20]. PI-RADS score 3 lesions are considered equivocal for PCa [21]. Whether this type of lesion needs further diagnostic workup remains controversial. Studies have shown that the prevalence of csPCa in PI-RADS score 3 lesions ranged from 10% to 30% (mostly around 20%) [20,22]. If prostate biopsies were performed in patients with PI-RADS scores of 3, there would be approximately 80% unnecessary aggressive procedures, and these procedures put the patients at risk. On the other hand, leaving them for observation may not be an acceptable choice because of the high missing rate (20%). Some studies showed that a lesion with a PI-RADS score of 3 was associated with a low likelihood of csPCa [23], and suggested that such lesions with a prior negative biopsy should not undergo a second biopsy [24]. Recently, the image texture analysis of T2WI and ADC maps of such lesions [25], as well as further classification of these patients was considered to reduce the missing rate. A meta-analysis showed that a threshold PSAD value ≥ 0.15 ng/mL² in these

cases is an indication for csPCa confirmation by combined systematic plus targeted biopsy [26]. Nomograms combining PI-RADS score, age, and PSAD were also developed for csPCa prediction in patients with non-suspicious or equivocal mpMRI result (i.e., PI-RADS score ≤ 3) [27]. The predictive performance of the combination of other biomarkers has also been studied [28]. PHI and PHID were also attempted to further classify patients with positive mpMRI results [29]. However, PHI and PHID have never been used for csPCa prediction in patients with PI-RADS scores ≤ 3 that were difficult to make decisions of further workup. In the present study, a threshold of PHI ≥ 51.84 or PHID ≥ 0.956 was discovered based on the Youden index with the same sensitivity of 85.00% for csPCa prediction. Regarding the higher AUC of the ROC curve (PHID: 0.829 [95% CI: 0.717–0.941] vs. PHI: 0.748 [95% CI: 0.610–0.886], $p < 0.05$) and relatively higher NPV of the cutoff value (PHID: 94.44% vs. PHI: 93.75%) than that of PHI, and an acceptable missing rate (i.e., false negative rate) of 15.00%, PHID was considered to be the best predictor of csPCa in mpMRI non-suspicious or equivocal patients with the highest benefit in DCA. Admittedly, the sensitivity (85.00%) of PHID threshold in patients with PI-RADS score 1 to 3 did not exceed PHI in those with gray-zone PSA level (sensitivity 88.64%; Table 3), but PHID did play an important role of csPCa prediction in patients with seemingly negative result on mpMRI, helping decision-making regarding biopsy in clinical setting.

There are several limitations to the present study. First, this single-centered small-sized cohort of Asian men was recruited from a tertiary medical center that may not be a good representation of the whole population. The cutoff value of PHI varied between Asian and European population [30] as reported. Second, the outcome of csPCa or PCa was based on systematic (69/222, 31.1%), targeted (4/222, 1.8%), and combined biopsies (149/222, 67.1%). As PHI and other biomarkers, and biochemical indexes were obtained for all the lesions, all positive cores were included (irrespective of the biopsy method). Biopsy-naïve (207/222, 93.2%) and repeated-biopsy (15/222, 6.8%) patients were also even-powered. Hence, further studies with larger populations and a PI-RADS v2.1 scoring system may be needed to confirm the improved performance of PHI and PHID for predicting csPCa.

CONCLUSIONS

In conclusion, PHI and PHID are useful predictors of csPCa. Not only can both serve as a triage tool for the whole population (as earlier reported) but they can also decrease the number of missed cases among patients with non-sus-

picious or equivocal MRI result (i.e., PI-RADS score ≤ 3). For this group of patients, a PHID ≥ 0.956 is associated with a higher likelihood of csPCa and is an indication for biopsy.

CONFLICTS OF INTEREST

The authors have nothing to disclose.

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AUTHORS' CONTRIBUTIONS

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SUPPLEMENTARY MATERIALS

Supplementary materials can be found via <https://doi.org/10.4111/icu.20230060>.

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